Persistent Homology for the Quantification of Microstructures in HRTEM Images

Abstract

Analysis of Transmission Electron Microscopy images provides a useful source of materials science research, but conventional methods are sensitive to imaging conditions and rely on manual processing or parameter-dependent techniques. This work proposes an unsupervised approach to crystal segmentation based on Persistent Homology, a topological data analysis technique that extracts atomic structural information at multiple scales by investigating image sub level sets. Topological signatures are extracted from an image and clustered to distinguish between crystalline and amorphous regions, with sub clusters further investigated through persistence density and image processing. Building on prior applications of Persistent Homology in electron microscopy, this work implements an application to crystal segmentation in High Resolution Transmission Electron Microscopy imaging, providing a structure-driven method for materials analysis.

1 Introduction

High Resolution Transmission Electron Microscopy (HRTEM) is a useful tool in investigating materials at the atomic level, but analysis is often subject to imaging conditions, manual processing, and interpreter bias [4]. Therefore, the ability to reliably extract useful information from images is vital. When differentiating between crystal and amorphous material, previous processing pipelines rely heavily on image processing and filtration methods, such as Fast Fourier Transform, to highlight specific variations in the image [6]. These methods are often subject to image conditions and so a new method is proposed that relies on image structure to identify crystalline regions - Persistent Homology (PH). Persistent Homology is a subfield of topological data analysis focused on identifying topological features. It does this by tracking how the topology of the image changes over increasing thresholds, allowing for the identification of structure based on feature lifetimes [2]. The benefit of PH is its unsupervised nature, removing the need for human-annotated training data and driven by the structure of the data. Previous works based on PH include image classification for amorphous states [5] and segmentation in alternate contexts such as medical image processing [1]or generic grayscale images [3]. These works illustrate the potential for automated classification and segmentation of HRTEM images. This work therefore aims to use PH in the identification of crystals in HRTEM images, as well as further analyse crystalline and amorphous structural signatures.

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2 Methods

In order to explore the effectiveness of persistent homology as an analysis tool for TEM images, this work utilises the HRTEM nanoparticle images dataset by Sytwu et al. [4] - images are processed as in Figure 1 using persistent homology and clustering. Here, an image is provided as an input and processed using a sliding window technique. A sublevel filtration is computed on each of these patches, resulting in a persistent homology representation - this representation is then vectorised using persistence images. These vectorised windows are processed using UMAP and a Gaussian Mixture Model (GMM) into clusters which are subsequently identified as crystalline or amorphous. Experiments are performed on a range of 4 to 8 clusters, allowing for the exploration of both broad classification and more detailed sub cluster analysis. This identification of structure is quantified and further explored using persistence density measures, allowing the recognition of specific structural topologies.

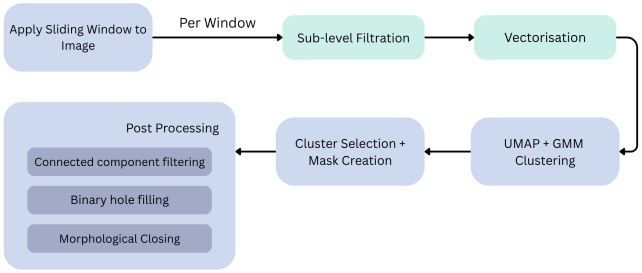


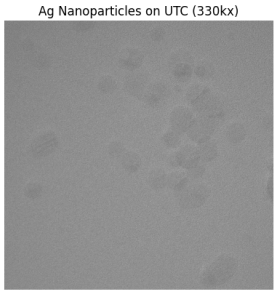
Figure 1: Pipeline for HRTEM image analysis: Image splitting, persistence feature extraction and clustering to extract mask of crystalline regions

3 Results

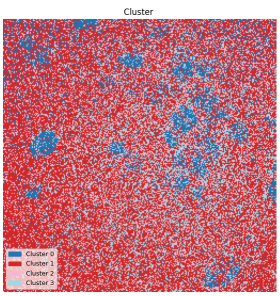
The images in Figure 2 are a product of the application of the pipeline in Figure 1 and illustrate the potential efficacy of persistent homology in crystal segmentation. When compared against human annotation labels, the mask achieves an IoU of 0.73, however visual analysis illustrates the granularity highlighted by this method - segmenting to a level of detail not typically afforded by manual annotation method annotations. A significant challenge is presented in the nature of the HRTEM images in terms of both imaging artifacts and salt and pepper noise, which is explored further through the application of both pre- and post-processing experiments.

These initial results also highlight the specific advantages of applying persistent homology for crystal analysis as we are able to investigate further exactly why these clusters emerged by examining their topological signatures. For example, the representative

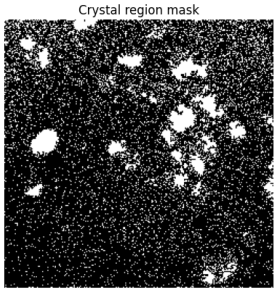
persistence diagrams in Figure 3 visually show how, even within the non-crystalline clusters, there are varying levels of similarity to crystalline structure, with cluster 1 exhibiting the clearest difference and making up the majority of amorphous regions. This is further confirmed by the persistence density plots in Figure 4, which illustrate how clusters 2 and 3 exhibit more similar distributions to cluster 0 (Crystalline), perhaps implying these structures share some underlying features with the crystalline regions. It is this style of analysis that drives the motivation to cluster in such detail (up to 8 clusters) as we are able to explore the signatures exhibited by such sub clusters. Additionally, a preliminary XGBoost model utilising label images was trained and tested on a subset of 13 images as a baseline, producing similar results.



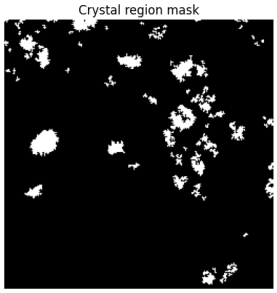
(a) Original HRTEM Image: Ag on UTC, 5nm nanoparticle diameters, 330kX



(b) Cluster map of vectorised persistences of 16x16 patches using UMAP & GMM, n = 4



(c) Crystalline cluster mask, segmented on cluster 0 (chosen manually)



(d) Mask after connected component filtering, binary hole filling and morphological closing

Figure 2: Results of crystalline clustering on HRTEM Image

4 Conclusions and Future Work

The proposed next step in segmentation is to train a model with the ability to segment full images using this persistent homology method. By utilising the clustered window data, a model may be trained to learn and classify certain topological signatures. Any test image therefore undergoes the same process of sliding window, sublevel filtration and vectorisation - the model then predicts, per patch, a crystallinity score which is used to reconstruct a segmentation mask.

The topological representations developed within this work are naturally suited to cross-material and cross-condition generalisation as they are based on the geometry of the image rather than learned from labels. This style of processing offers an interesting direction for materials modelling in improving transfer across sample types and imaging parameters. More broadly, persistent homology also provides a form of structural explanation in which AI-driven segmentation can be justified in terms of specific topological features. This would allow an AI-based system to provide interpretation to users in terms of topological structure – both in terms of what topology a material exhibits and how that factors into classification. This points towards a wider field of investigation into AI for unsupervised materials analysis, allowing broader data analysis that is unrestricted by the availability of human annotations.

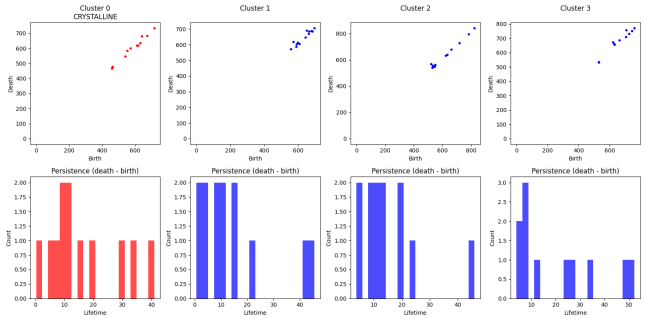


Figure 3: Clustering Analysis - Representative Persistence Diagram per cluster

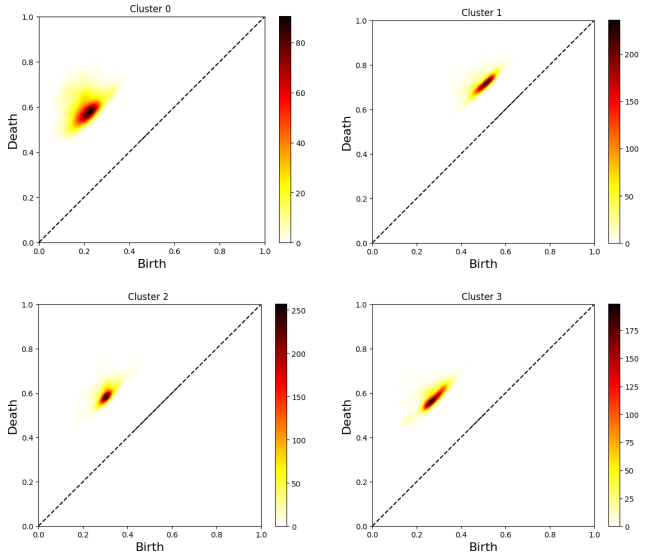


Figure 4: Clustering Analysis - Persistence Density Diagram per cluster

Acknowledgments

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